

# Kvinnobion

from the KvinnSam archives



Digitalisering för bevarande och tillgänglighet C3IDB1

B11V6 VT2026

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# Introduction

## Background

The women's movement in both a global and a national context is an important part of our common cultural heritage. Historically it has been important for creating the gender equality we have today in Sweden. Kvinnobion, which was an activity group at one of Sweden's first women's houses Kvinnohuset, was part of the women's movement during the late 1970s and beginning of the 1980s and formed a creative hub where women gathered to watch and talk about movies and movie making with a focus on movies made by women, about women.

## The source material

The material for this project consists mainly of movie schedules from Kvinnobion, posters presenting individual movies and posters and folders from movie festivals that Kvinnobion was involved in. This material gives an interesting perspective of what movies were made, watched and discussed in relation to the women's movement and feminism during the late 1970s and the 1980s.

All the graphical designs for posters and movie schedules were made by a woman named Marie Falksten, who was actively involved in Kvinnobion as well as other women's movement activities at Kvinnohuset. Marie Falksten has also written most of the texts on these movie schedules and posters.

## Relevance for cultural heritage preservation

Sula (2015, p. 13-36) describes how the digitization of cultural heritages opens up the possibilities of new ways of interacting with and studying history and culture. With the use of digital tools and resources, digitized cultural heritage can give many different people access to material and enable new forms of interdisciplinary study. Kvinnobion is a fascinating historical artifact that intersects several possible fields of research. It is a part of a vibrant piece of local history, of cultural organizations and practices; perspectives on LGBTQ life and community. There is also a lot of material detailing individual cultural production through the years. Our goal with this project is to preserve this material and to make it available for those who wish to study it, or otherwise take part in its history. To this end we wanted to focus on presenting the graphic and textual contents of the artifacts.

## Method

### Selection

We referred to Cowick's (2018, p. 30-39) guidelines as a starting point for making our selection. Cowick presents five criteria for selections: 1) Value; 2) Demand; 3) Condition of

materials; 4) Legal and ethical issues; 5) Availability. Although some of these criteria were not as relevant to us as others, they provided a solid standard for us to shape our selection.

Selection of the material was done over two occasions (17 and 23 March). The selection process was carried out in close communication with Rachel Pierce, our contact at Kvinnsam, who helped specify which parts of Kvinnsam's archives could be of interest. From the specified collections several boxes were ordered. The first round of selections was used to sort through the materials in the boxes, to make note of possible objects to digitize and identify objects that would have to be ignored for legal reasons.

Rachel was then notified of which objects we were interested in. In the second round, twenty-one objects were selected from the remaining materials. Three of these were then discarded for a final eighteen objects.

## Copyright and legal aspects

The primary legal aspects affecting our selection were issues of copyright and personal data. Our final selection was made up of graphic designs created by Marie Falksten, who provided a license to Kvinnsam to use this work for digitization. This resolved the issues regarding ownership. We tried to limit any materials containing personal information. The treatment of these articles will be discussed in the ethics section. There were also images containing the logo for Hagabion, which we contacted in order to gain permission to use for the reproductions.

## Image capturing and editing

In preparation for the image capture we used the DFG Practical Guidelines (Deutsche Forschungsgemeinschaft, 2013) and Kungliga bibliotekets instruktioner (Kungliga biblioteket, 2011) as a baseline to gain an understanding of what we would need for the image capture and to construct a workflow for the image processing.

The image capture was carried out on 2 April in one of Kvinnsam's offices at the University of Gothenburg's library using a camera (technical specifications can be found in the appendix) as we did not have access to a scanner. We wore neutral-colored clothing so as not to distort the colors. We also used gray paper as a background for our photos. A tripod was used to stabilize the camera. The tripod didn't allow the camera to be positioned to shoot downwards, but an extension arm was built to solve this problem. The camera was connected to a laptop, so we could review the photos on-site. A remote trigger was used to take the pictures.

Image processing was done in GIMP (version 3.1.2) using the extension RawTherapee (version 5.12) to allow working with the camera raw-files. As our objective with the digitization primarily was to reproduce and preserve the semiotic contents of the artifacts, we chose to only capture both sides of objects that had writing or images on the backside. For blank backs only the fronts were captured. The images were edited to correct for visual noise and sharpness.

## Text capturing and TEI

As a camera was used for the image capture and most of our digitized objects only contained a limited amount of text, we decided to transcribe the text manually instead of using OCR. According to the DFG Practical Guidelines (Deutsche Forschungsgemeinschaft, 2013, p. 33) the accuracy is often higher for manually transcribed text. For the transcription we chose a single-key method with one person transcribing and a second person proofreading the transcriptions. These transcriptions then formed the basis for the text encoding according to TEI.

Dahlström (2009, p. 175) argues that there are different levels of transcription, for instance a strictly faithful or a freer transcription. We have chosen to be faithful to the original by reproducing typos from the original in the transcribed text. During the text encoding these were marked up with TEI elements providing both the original and the corrected version. We also presented these typos and incorrect wordings, and their corrections, on the website. We wanted the digitized version to stay faithful to the original, while showing that corrections had been made for some words.

After transcription, the text was marked up and metadata added with the TEI markup language according to the recommendations in the DFG Practical Guidelines (Deutsche Forschungsgemeinschaft, 2013, p. 8). We took inspiration from exercises done on campus at the University of Borås and used the TEI Guidelines (The TEI Consortium, 2026) to construct the TEI code through an iterative process, which included choosing the TEI elements we deemed most suitable for describing our digitized objects. For the overall structure we had two alternatives: either to construct multiple separate TEI files or to group texts that were related into a TEI corpus. We chose to group a selection of movie schedules and posters from Kvinnobion with a lot of metadata in common into one TEI file with a common TEI corpus. Some of the digitized objects didn't fit into this first grouping though and were therefore described in individual TEI documents. All TEI code was proofread by a second person.

## Metadata

Dappert et al. (2016, p. 3) points out that it is important to adhere to standards for metadata as they contribute to best practices and facilitate the metadata management. We have chosen to follow the Dublin Core Metadata Initiative for embedded metadata in image files. We also provided metadata by encoding the text according to the Text Encoding Initiative (TEI) guidelines. Furthermore we have added keywords from Kvinnsam's subject headings as well as concepts from Queerlit's thesaurus that uses linked data and maps to the English thesaurus Homosaurus. In the TEI files we added descriptive, structural and administrative metadata. A `xenodata` element referring to technical metadata embedded in the image files was also added.

Metadata was embedded in the images using GIMP. Titles, descriptors and creator notes were added to each image, along with rights statements for legal reasons. Metadata was then extracted from the files using the software Exiftool and written to txt-files (complications with Exiftool changed our approach to editing and delivering this data, which will be discussed below).

We have chosen to write our metadata in Swedish as both the context and our primary target audience are Swedish.

## XSL transformation and publication

We based the XSL transformation on an XML template that was shared with us as part of the course in which the project is included. We made adjustments to the template so that it would generate an HTML-structure according to how we thought it would be most suitable to communicate our digitized objects on our website. For instance we chose the structure of showing one object and its facsimiles at a time, together with the associated transcriptions (instead of showing one page with several digitized objects together). Our intention was to let the viewer focus on one object at a time and really get the chance to dive into it. The concept of “diving into it” was further enhanced through a small JavaScript function that we coded to construct a magnifying glass that is activated when one hovers over a facsimile on the webpage. At the same time we do not have control over how the actual users of the website will interpret these choices that we made concerning the presentation. Like Björk (2015, p. 2) describes it, the digitization on a conceptual level puts interpretations and boundaries of objects in motion in a much less controlled way than in the physical/material world. Our digitized representation is influenced by the institutional practices from which we gathered the actual historical objects, Kvinnsam. How we have chosen to present the digitized version is also influenced by this institutional context (Björk, 2015, p.12-13).

## Long-term preservation

The photographs were saved in the RW2 file format, which is a raw image format used by Panasonic cameras. The raw images were then used to create the masterfiles of the unedited images to be saved for preservation. These masterfiles were saved in the lossless TIFF format. Images were then processed for publication and saved as jpg files.

For the sake of long-term preservation the TIFF versions of the masterfiles were saved and made available through a google drive account dedicated to the project (with the exception of files that contained sensitive personal information).

We also store copies of our digital masterfiles and our TEI files on two different external hard drives, which according to Cowick (2018, p. 44) is the basic level for preserving digital items, suitable for small budgets.

## Analysis and discussion

### Ethical aspects

Manžuch (2017, p. 7-14) writes about the possible ethical issues inherent in the digitization of cultural heritage, that there is a risk of infringing the privacy of groups or individuals. This

was something we were wary of during our development of this project. During our first round of selections we found several objects, mainly logbooks and organizational diaries, that were of interest to us. As a historic material they gave an incredible viewpoint on local history and of a subculture and their practices. However, due to the close and personal nature of these documents we were unable to use them. Navigating the legal issues of safeguarding the privacy of individuals and also sorting out the actual ownership of the documents, would be much too complex.

As mentioned above our selections were limited to materials created, and approved for digitization, by Marie Falksten. However there were still some pieces of personal information detailed on the objects, specifically the names of people to contact in Kvinnobion. Because we could not ascertain whether the persons in question were available for contact or still alive, we had to find different solutions to enable publication of the materials. We consulted our contact Rachel, as well as the literature by Manžuch, and arrived at a conclusion. We decided to censor the names in all the files to be published. Also we will not make the uncensored masterfiles available for public viewing, but we will keep them saved locally for future publication.

## Users and availability

This project was made for scholars of and those interested in women's history and to be a gateway for deeper study. Our selection had resulted in a decently sized amount of material with a wide variety of informational contents. We wanted to make this information available for viewing or study in full. To do this our aim was to present the materials in several different ways: images and text transcripts, with complete XML markup and embedded image metadata. Milekic (2007, p. 374-385) writes about how digitization can create tools that enable new ways of study and interpretation that are not possible with physical objects. By making objects available in different formats new ways of approaching that material is made possible.

We wanted to present the material in a way that enables this kind of multi-modal study. With complete images, text and metadata. The textual elements, for example, were fully transcribed and presented with complete markup that details the various signs, diacritic marks and different languages used in full. Metadata was linked to each object in order to detail authorship, provenance and more. This data was to be made available in both written text and embedded in the images (although due to technical difficulties we were unable to complete this aspiration).

## Critical digitization

Dahlström (2009, p. 176-183; 2011, p. 94-100) writes about mass digitization – focusing on quantity – versus critical digitization – focusing on quality – as the two ends of a scale. Our project is situated near the critical end of the scale with high ambitions for both text-oriented digitization and image-oriented digitization as well as metadata. As we have been working with a limited selection of objects, we have been able to focus on quality throughout the process. However, this being a student project, limited access to technical equipment, time constraints and lack of experience hampered our work. Our objects being archival material

has of course also meant that we have had to be very careful with it, and using automated methods from mass digitization (Dahlström, 2009, p. 177) would not have been an option. According to Dahlström (2009, p. 181) it's quite common that critical digitization projects only focus on facsimiles. However, we chose to render both the facsimiles and machine-readable, searchable transcribed texts.

## The importance of context

McCarthy (2007, p. 246-247) highlights the importance of systemically preserving contextual information about digital cultural heritage resources to prevent what he calls “epistemic failure” (McCarthy, 2007, p. 246), i.e. a lack of necessary information to be able “to explain the structure and meaning of the data” (McCarthy, 2007, p. 246). By documenting and preserving contextual information he states that we can make sure that this information doesn't get lost for future generations.

In line with this, we wanted to put our digitized material from Kvinnobion in context and create a narrative about it. This influenced our choices both at the selection level, the metadata level and the presentation level. More specifically, we chose to include a poster from Kvinnohuset and a poster from Biofem in our selection to provide some context, we included detailed metadata in the TEI headers and we also chose to tell the story of Kvinnohuset, including its relations to the above mentioned Kvinnohuset and Biofem, in a presentation text on our website. Hopefully this context can help future audiences of our digitized material to better understand it and thus facilitate its reuse.

## Challenges and limitations

### Image processing

From the start we were well aware that we would not meet several of the standards proposed in these documents. Our image capture would both be limited by our lack of technical equipment and the inability to control the lighting during the capture process. The fact that we did not have access to a color chart also meant we would not be able to accurately calibrate the colors during image processing. We decided on trying to keep any edits to the images to a minimum, in order not to over-correct the images.

Our approach to these issues was to make note of any deficiencies and try to work around them. We documented the nature of the environment during the image capture, so we would have a record of ambient lighting and other effects that could change the qualities of the photos. The primary source of light was from outside a window and we moved our photographing station as close as we could to get the best possible exposure, but the images were somewhat unevenly lit. Luckily the weather was in our favor as the day was overcast but light, so the lighting was neutral and quite good. However there was also some yellow light streaming in from a window connected to the corridor outside. At this point some more problems revealed themselves. First was the fact that the lowest ISO we could use on the camera was 200, and this resulted in a slight amount of image noise on all the pictures. Secondly, the highest resolution images that could be extracted from the raw files were at a

resolution of 180 dpi, which is (according to the standards we were working with) adequate for images meant for publishing but quite a bit lower than the 300 dpi that is recommended for masterfiles. There was nothing we could do about this. Then there were some issues that affected specific images. A couple of images had been affected by the light from the corridor and now showed a light orange hue (notable in large black portions of the image). These images received additional corrections to the hue of the colors.

Throughout the project we used a Github repository to save and share our work. Due to the large size of the TIFF files these could not be uploaded to the repository. To solve this problem we created secondary copies of the masterfiles in J2 format (jpeg2000) specifically for online publication.

## Text capturing

One of our objects contained a paragraph of text in Greek which caused some problems during text capturing as we needed to use the Greek alphabet. For this paragraph we decided to try out OCR by uploading the image file to Google Drive and opening it in Google Docs. This worked quite well, and we only needed to make minor corrections to the text. This text was then proofread by two different persons, one of whom had some knowledge of Ancient Greek that came in handy.

## Metadata

Extracting and editing the image metadata using Exiftool was a challenge. Extracting the data from the images was no problem as Exiftool incorporates a drag-and-drop function, where an image can be placed on top of the executable in order to extract its metadata, as blocks of text. However, we were unable to edit the metadata inside the image files themselves, due to some error with Exiftool that we were unable to correct. This meant that there would be some differences between the metadata presented in the images and the text files. Because of these challenges, we also had to make do with delivering the image metadata in the form of txt files instead of XML files. Ideally the image metadata would have been added directly in the xenodata element in the TEI markup, but since we didn't manage to extract it in the correct XML format, we added links to the text files containing the image metadata.

The txt files were edited so that the metadata added through GIMP would conform to the Dublin Core standard. Some irrelevant metadata was also erased. GIMP's metadata also showed a few quirks that were corrected: the creator-tag would sometimes duplicate itself and repeat its information. None of these corrections could unfortunately be made to the metadata inside the master and publication files.

As for the text metadata in TEI there were also some challenges regarding the choice of most suitable elements to use. The TEI guidelines describe a lot of different elements to use and even though we reflected on and discussed in the group which elements would be most suitable, there would of course be room for further improvements.



## XSL transformation

Since XSL transformation was not the focus of the course it was a bit challenging to make enough time to learn this more thoroughly. We did however manage to make adjustments to the template that we had access to and construct a webpage which is ok as a first version. To map the TEI structure to XSL code and get hold of the right elements through the XSL code (to be able to construct the HTML structure we had in mind) was somewhat challenging since XSL functions differently than other programming languages that the group had knowledge of since before. When working with the XSL code we discovered that some of the TEI code was not optimally structured according to the HTML we wanted to generate. We did some adjustments in the TEI code and in the XSL code and in the end managed to work around some of these issues, and for the most part correctly display columns with text where this was needed. There are still some parts of the transcribed texts that are not displayed exactly as in the facsimiles but to fix it would require quite much testing with HTML and CSS, which we didn't have enough time for within the project boundaries. A limitation in the current implementation of the XSL transformation is that we have a lot of duplicated code, which means it is not very maintainable. Ideally one would have one XSL file that generates similar HTML pages, and which takes an XML id (pointing to the individual TEI object within as in our case the TEI corpus) as a parameter and then generates different HTML pages depending on the value of the ID. Because of time and scope limits this was not fixed in this version of the project.

## Conclusions

Rachel provided us with valuable feedback on the project. Specifically on how we could make the website more user-friendly by incorporating links to Kvinnsams archives with the digital objects and by clarifying how we wanted the site to be used. She also helped us highlight the international and historic context in which Kvinnobion existed. We received this feedback quite late but nevertheless managed to integrate most of the suggested changes.

## Scalability and recommendations for future work

This being a small-scale student project has of course entailed some limitations. According to Dahlström (2009, p. 181) critical digitization projects risk being forgotten and thus one possible continuation of our project could be to put some effort into marketing it, for example through social media. Another possibility could be to translate our website into English to be able to reach a wider audience.

The large amount of material available through Kvinnsam's archives could make it possible to expand this project in several ways. The material we have published could be complemented by further digitization efforts aimed towards any of Kvinnobion's remaining objects. If a project had the means to resolve the legal aspects inherent to some of the objects, it could expand the scope of the project greatly. This would give the project a much greater depth and enable more granular levels of historic study.

As the archives also contain many items that are related to Kvinnobion without being directly part of it, there also exists the possibility of future projects that intersect with this one and in turn add to its context. There is much material in the archives belonging to Biofem for example, or objects that touch on Swedish feminist film production.

From a technical perspective it would be recommended to do further work on the structure of the website and the handling of the digitized items. If more time was available, one would surely want to have a more rigid backend solution for storage of transcribed texts and images. This backend would likely consist of some sort of SQL database, as well as a code layer handling the logic of the website, and in addition to that this would interact with the frontend code that we already provided in this project. To separate storage, logic and presentation into different layers would make the website more maintainable in a long-term perspective and would also mean one followed best practices for web development. In addition more work on the frontend of the webpage would also have to be done, there are for example still things that would need to be fixed with CSS styling to make for example the transcribed texts even more identical in their layout to the actual facsimiles. To enhance the value of the website it would be advisable to create a search function that would make it possible for users to search for different digitized objects by typing in keywords that would match against the metadata keywords that we added in our TEI. Also it would be interesting if one could make the digitized version of the objects even more interactive for users and make abstract information more tangible and easily accessible like how Milekic discusses in his article (Milekic, 2007).

As we were unable to complete the embedding and editing of metadata in the publication and masterfiles, this would also be a clear subject for future work.

## Key learnings

This project has put into sharp focus the high levels of specialized labor and technical resources needed to perform any digitization effort. Different kinds of materials require wildly different modes of processing, both to capture the object and to render and present the information in a useful way. We were fairly well equipped with prior knowledge and experience to make this project happen and we still had to learn on the fly. If our aim had been to create a project at the levels of standard put forward by the literature the level of complexity would have increased manifold. This project has given us a broader perspective on the width of the field digitized cultural heritage. The amounts of materials that could be made available for digitization are enormous, and so is also the potential required labor. This work is of course compounded by the intricacies of the legal aspects affecting these artifacts.

This complexity has also made working on this project very rewarding. It has been an enormous opportunity to collaborate and learn new skills. By engrossing ourselves in this material we have been able to take part in this fascinating piece of history. Kvinnobion is a very interesting part of the cultural heritage of Gothenburg, and of feminist and LGBTQ history. It has been a pleasure to be part of the work to preserve and make this material available to more people.

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# Appendix A

## Time spent

| Activity                    | Hours spent |
|-----------------------------|-------------|
| Planning and meetings       | 70          |
| Reading relevant literature | 120         |
| Selection                   | 22          |
| Image capture               | 22          |
| Image editing               | 29          |
| Text capture                | 15          |
| Metadata                    | 33          |
| TEI                         | 81          |
| HTML, XSLT and website      | 63          |
| Project report              | 56          |
| <b>Total hours spent</b>    | <b>511</b>  |

The table above shows an estimation of time spent on different activities within the project. Much time was spent on reading relevant literature before and during the project, and text encoding according to TEI also proved to be a time-consuming task. The number of hours for TEI encoding includes preparations, the encoding as such and proofreading of the code. The selection also took more time than anticipated as we needed to return to Kvinnsam a second time to complete the selection.

The table below shows a very rough time estimate for different possible continuations of the project mentioned in the report. The allocated time would of course depend on a number of factors, such as the scale, the long-term goal of the project, unexpected events et cetera.

| Possible continuations    | Time estimate (hours) |
|---------------------------|-----------------------|
| Marketing, initial effort | 20                    |
| English site              | 30                    |
| Adding more material      | 30                    |
| Backend solutions         | 200-400               |
| Embedding of metadata     | 12                    |

# Appendix B

## Hardware and Software – Technical Specifications

### Camera

Panasonic Lumix DMC-GX8

Exposure time: 1/100

ISO: 200

Shutter speed: 1/100

Light value: 9.8

### Lens

LUMIX G VARIO 14-140/F3.5-5.6

Focal length: 26.0 mm

### Software

GIMP 3.2.0

Color profile: sRGB

Resolution: 180 DPI

RawTherapee 5.12

Oxygen XML Editor 28.0

Exiftool 13.57